

Project 6 – Library Management System

Create table Books and Members and insert 20 records.

The screenshot shows the SQL Developer interface with the following SQL script:

```
57 INSERT INTO books VALUES (120, 'AI Fundamentals', 'Andrew Ng', 'Programming', 1800, 2025, 27);
58
59 select * from books;
60
61 select * from members;
```

The Results pane displays the data for the 'books' and 'members' tables.

book_id	title	author	category	price	publication_year	copies
101	Learning SQL	Alan Beaulieu	Programming	650.00	2022	25
102	Python Basics	John Smith	Programming	750.00	2023	18
103	Java Complete Guide	Herbert Schildt	Programming	850.00	2021	12
104	C Programming	Dennis Ritchie	Programming	550.00	2020	5
105	Data Structures	Narasimha Karumanchi	Programming	900.00	2024	30
106	Database Systems	Elmasri	Programming	1200.00	2025	15
107	Operating Systems	Galvin	Programming	950.00	2021	10
108	Computer Networks	Tanenbaum	Programming	1000.00	2020	0

member_id	member_name	city
1	Arun	Chennai
2	Priya	Madurai
3	Deepa	Chennai
4	Raj	Salem
5	Karthik	Trichy
6	Suresh	Coimbatore
7	Swetha	Madurai

Query executed successfully.

Display books with price greater than 500

The screenshot shows the SQL Developer interface with the following SQL script:

```
57 INSERT INTO books VALUES (120, 'AI Fundamentals', 'Andrew Ng', 'Programming', 1800, 2025, 27);
58
59 select * from books;
60
61 select * from members;
62
63 select * from books where price >=500;
```

The Results pane displays the data for the 'books' table, filtered by price greater than or equal to 500.

book_id	title	author	category	price	publication_year	copies
101	Learning SQL	Alan Beaulieu	Programming	650.00	2022	25
102	Python Basics	John Smith	Programming	750.00	2023	18
103	Java Complete Guide	Herbert Schildt	Programming	850.00	2021	12
105	Data Structures	Narasimha Karumanchi	Programming	900.00	2024	30
106	Database Systems	Elmasri	Programming	1200.00	2025	15
107	Operating Systems	Galvin	Programming	950.00	2021	10
108	Computer Networks	Tanenbaum	Programming	1000.00	2020	0
109	SQL Cookbook	Anthony Molinaro	Programming	700.00	2026	8
113	Atomic Habits	James Clear	Self Help	800.00	2023	28
114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17
116	Clean Code	Robert C. Martin	Programming	950.00	2024	20
117	SQL in 10 Minutes	Ben Forta	Programming	680.00	2025	6
118	Power BI Guide	Raj Kumar	Programming	780.00	2026	0
119	Machine Learning	Tom Mitchell	Programming	1500.00	2026	35
120	AI Fundamentals	Andrew Ng	Programming	1800.00	2025	27

Query executed successfully.

Display books in the programming category

```
57      INSERT INTO books VALUES (120, 'AI Fundamentals', 'Andrew Ng', 'Programming', 1800, 2025, 27);
58
59      select * from books;
60
61      select * from members;
62
63      select * from books where price >=650;
64
65      select * from books where category='programming';
66
67
68
69
70
71
72
73
74
```

100 % ● No issues found Ln: 65, Ch: 1 (49 chars, 1 line)

Results Messages

	book_id	title	author	category	price	publication_year	copies
1	101	Learning SQL	Alan Beaulieu	Programming	650.00	2022	25
2	102	Python Basics	John Smith	Programming	750.00	2023	18
3	103	Java Complete Guide	Herbert Schildt	Programming	850.00	2021	12
4	104	C Programming	Dennis Ritchie	Programming	550.00	2020	5
5	105	Data Structures	Narasimha Karumanchi	Programming	900.00	2024	30
6	106	Database Systems	Elmasri	Programming	1200.00	2025	15
7	107	Operating Systems	Galvin	Programming	950.00	2021	10
8	108	Computer Networks	Tanenbaum	Programming	1000.00	2020	0
9	109	SQL Cookbook	Anthony Molinaro	Programming	700.00	2026	8
10	110	HTML & CSS	Jon Duckett	Programming	600.00	2022	22
11	116	Clean Code	Robert C. Martin	Programming	950.00	2024	20
12	117	SQL in 10 Minutes	Ben Forta	Programming	680.00	2025	6
13	118	Power BI Guide	Raj Kumar	Programming	780.00	2026	0
14	119	Machine Learning	Tom Mitchell	Programming	1500.00	2026	35
15	120	AI Fundamentals	Andrew Ng	Programming	1800.00	2025	27

Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row

Display books published between 2020 and 2026

```
58      select * from books;
59
60      select * from members;
61
62
63      select * from books where price >=650;
64
65      select * from books where category='programming';
66
67      select * from books where publication_year between 2020 and 2026;
68
69
70
71
72
73
74
```

100 % ● No issues found Ln: 67, Ch: 1 (65 chars, 1 lines) T

Results Messages

	book_id	title	author	category	price	publication_year	copies
1	101	Learning SQL	Alan Beaulieu	Programming	650.00	2022	25
2	102	Python Basics	John Smith	Programming	750.00	2023	18
3	103	Java Complete Guide	Herbert Schildt	Programming	850.00	2021	12
4	104	C Programming	Dennis Ritchie	Programming	550.00	2020	5
5	105	Data Structures	Narasimha Karumanchi	Programming	900.00	2024	30
6	106	Database Systems	Elmasri	Programming	1200.00	2025	15
7	107	Operating Systems	Galvin	Programming	950.00	2021	10
8	108	Computer Networks	Tanenbaum	Programming	1000.00	2020	0
9	109	SQL Cookbook	Anthony Molinaro	Programming	700.00	2026	8
10	110	HTML & CSS	Jon Duckett	Programming	600.00	2022	22
11	111	The Alchemist	Paulo Coelho	Novel	450.00	2020	14
12	112	Wings of Fire	A.P.J Abdul Kalam	Biography	350.00	2021	40
13	113	Atomic Habits	James Clear	Self Help	800.00	2023	28
14	114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17
15	115	Think and Grow Rich	Napoleon Hill	Finance	500.00	2021	9
16	116	Clean Code	Robert C. Martin	Programming	950.00	2024	20
17	117	SQL in 10 Minutes	Ben Forta	Programming	680.00	2025	6

Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, C

Select R

Display books written by a specific author

The screenshot shows a SQL IDE with a query editor and a results pane. The query editor contains the following SQL code:

```
59 select * from books;
60
61 select * from members;
62
63 select * from books where price >=650;
64
65 select * from books where category='programming';
66
67 select * from books where publication_year between 2020 and 2026;
68
69 select * from books where author='Galvin';
70 select * from books where author!='Galvin';
71 select * from books where author<>'Galvin';
72 select * from books where author in ('Galvin','James clear');
73 select * from books where author='Galvin' or author='James clear';
74 select * from books where copies <=20;
75 select author from books where publication_year >=2020;
76
```

The results pane shows a table with one column, 'author', and 17 rows of data:

author
1 Alan Beaulieu
2 John Smith
3 Herbert Schildt
4 Dennis Ritchie
5 Narasimha Karumanchi
6 Elmasri
7 Galvin
8 Tanenbaum
9 Anthony Molinaro
10 Jon Duckett
11 Paulo Coelho
12 A.P.J Abdul Kalam
13 James Clear
14 Robert Kiyosaki
15 Napoleon Hill
16 Robert C. Martin
17 Ben Forta

The status bar at the bottom indicates: "Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Col: 1".

Display book titles containing SQL

The screenshot shows a SQL IDE with a query editor and a results pane. The query editor contains the following SQL code:

```
73 select * from books where author='Galvin' or author='James clear';
74 select * from books where copies <=20;
75 select author from books where publication_year >=2020;
76
77 select * from books;
78
79 select * from books where title like '%SQL%';
80
81
82
83
84
85
86
87
88
89
90
```

The results pane shows a table with 7 columns: 'book_id', 'title', 'author', 'category', 'price', 'publication_year', and 'copies'. It contains 3 rows of data:

book_id	title	author	category	price	publication_year	copies
1 101	Learning SQL	Alan Beaulieu	Programming	650.00	2022	25
2 109	SQL Cookbook	Anthony Molinaro	Programming	700.00	2026	8
3 117	SQL in 10 Minutes	Ben Forta	Programming	680.00	2025	6

The status bar at the bottom indicates: "Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Col: 1".

Update the price of all programming books by 10%

The screenshot shows the SQL Developer interface with a query window titled 'SQLQuery5.s...deepa (77)*'. The query contains several SELECT statements and one UPDATE statement. The UPDATE statement is highlighted in blue: `update books set price = price + (price*0.15) where category = 'programming';`. The status bar at the bottom indicates 'Query executed successfully.' and 'DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, C'.

```
73 select * from books where author='Galvin' or author='James clear';
74 select * from books where copies <=20;
75 select author from books where publication_year >=2020;
76
77 select * from books;
78
79 select * from books where title like '%SQL%';
80
81 update books set price = price + (price*0.15) where category = 'programming';
82
83
84
85
86
87
88
89
90
```

book_id	title	author	category	price	publication_year	copies
101	Learning SQL	Alan Beaulieu	Programming	859.63	2022	25
102	Python Basics	John Smith	Programming	991.88	2023	18
103	Java Complete Guide	Herbert Schildt	Programming	1124.13	2021	12
104	C Programming	Dennis Ritchie	Programming	727.38	2020	5
105	Data Structures	Narasimha Karumanchi	Programming	1190.25	2024	30
106	Database Systems	Elmasri	Programming	1587.00	2025	15
107	Operating Systems	Galvin	Programming	1256.38	2021	10
108	Computer Networks	Tanenbaum	Programming	1322.50	2020	0
109	SQL Cookbook	Anthony Molinaro	Programming	925.75	2026	8
110	HTML & CSS	Jon Duckett	Programming	793.50	2022	22
111	The Alchemist	Paulo Coelho	Novel	450.00	2020	14
112	Wings of Fire	A.P.J Abdul Kalam	Biography	350.00	2021	40
113	Atomic Habits	James Clear	Self Help	800.00	2023	28
114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17
115	Think and Grow Rich	Napoleon Hill	Finance	500.00	2021	9
116	Clean Code	Robert C. Martin	Programming	1256.38	2024	20
117	SQL in 10 Minutes	Ben Forta	Programming	899.30	2025	6

Delete books with zero copies available

The screenshot shows the SQL Developer interface with a query window titled 'SQLQuery5.s...deepa (77)*'. The query contains several SELECT statements and one DELETE statement. The DELETE statement is highlighted in blue: `Delete books where copies=0;`. The status bar at the bottom indicates 'Query executed successfully.' and 'DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Col: 1 18 rows'.

```
73 select * from books where author='Galvin' or author='James clear';
74 select * from books where copies <=20;
75 select author from books where publication_year >=2020;
76
77 select * from books;
78
79 select * from books where title like '%SQL%';
80
81 update books set price = price + (price*0.15) where category = 'programming';
82
83 Delete books where copies=0;
84
85
86
87
88
89
90
```

book_id	title	author	category	price	publication_year	copies
101	Learning SQL	Alan Beaulieu	Programming	859.63	2022	25
102	Python Basics	John Smith	Programming	991.88	2023	18
103	Java Complete Guide	Herbert Schildt	Programming	1124.13	2021	12
104	C Programming	Dennis Ritchie	Programming	727.38	2020	5
105	Data Structures	Narasimha Karumanchi	Programming	1190.25	2024	30
106	Database Systems	Elmasri	Programming	1587.00	2025	15
107	Operating Systems	Galvin	Programming	1256.38	2021	10
109	SQL Cookbook	Anthony Molinaro	Programming	925.75	2026	8
110	HTML & CSS	Jon Duckett	Programming	793.50	2022	22
111	The Alchemist	Paulo Coelho	Novel	450.00	2020	14
112	Wings of Fire	A.P.J Abdul Kalam	Biography	350.00	2021	40
113	Atomic Habits	James Clear	Self Help	800.00	2023	28
114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17
115	Think and Grow Rich	Napoleon Hill	Finance	500.00	2021	9
116	Clean Code	Robert C. Martin	Programming	1256.38	2024	20
117	SQL in 10 Minutes	Ben Forta	Programming	899.30	2025	6
118	Machine Learning	Tom Mitchell	Programming	1083.75	2026	36

Display books status using case copies >20 as available, copies >5 as limited else out of stock

SQLQuery5.s...deepa (77)* SQL project 5...A\deepa (75)

```
75 select author from books where publication_year >=2020;
76
77 select * from books;
78
79 select * from books where title like '%SQL%';
80
81 update books set price = price + (price*0.15) where category = 'programming';
82
83 delete books where copies=0;
84
85 select *,
86 case
87 when copies>=20 then 'Available'
88 when copies>=5 then 'Limited'
89 else 'Out of Stock'
90 end as Copies_Current_Status
91 from books;
92
```

100 % No issues found Ln: 84, Ch: 1 (139 chars, 8 lines) TAF

Results Messages

	book_id	title	author	category	price	publication_year	copies	Copies_Current_Status
1	101	Learning SQL	Alan Beaulieu	Programming	859.63	2022	25	Available
2	102	Python Basics	John Smith	Programming	991.88	2023	18	Limited
3	103	Java Complete Guide	Herbert Schildt	Programming	1124.13	2021	12	Limited
4	104	C Programming	Dennis Ritchie	Programming	727.38	2020	5	Limited
5	105	Data Structures	Narasimha Karumanchi	Programming	1190.25	2024	30	Available
6	106	Database Systems	Elmasri	Programming	1587.00	2025	15	Limited
7	107	Operating Systems	Galvin	Programming	1256.38	2021	10	Limited
8	109	SQL Cookbook	Anthony Molinaro	Programming	925.75	2026	8	Limited
9	110	HTML & CSS	Jon Duckett	Programming	793.50	2022	22	Available
10	111	The Alchemist	Paulo Coelho	Novel	450.00	2020	14	Limited
11	112	Wings of Fire	A.P.J Abdul Kalam	Biography	350.00	2021	40	Available
12	113	Atomic Habits	James Clear	Self Help	800.00	2023	28	Available
13	114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17	Limited
14	115	Think and Grow Rich	Napoleon Hill	Finance	500.00	2021	9	Limited
15	116	Clean Code	Robert C. Martin	Programming	1256.38	2024	20	Available
16	117	SQL in 10 Minutes	Ben Forta	Programming	899.30	2025	6	Limited
17	119	Machine Learning	Tom Mitchell	Programming	1983.75	2026	35	Available

Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Co

Display books ordered by publication year descending

SQLQuery5.s...deepa (77)* SQL project 5...A\deepa (75)

```
80
81 update books set price = price + (price*0.15) where category = 'programming';
82
83 delete books where copies=0;
84
85 select *,
86 case
87 when copies>=20 then 'Available'
88 when copies>=5 then 'Limited'
89 else 'Out of Stock'
90 end as Copies_Current_Status
91 from books;
92
93 select * from books order by publication_year desc;
94
95
96
97
```

100 % No issues found Ln: 93, Ch: 1 (51 chars, 1 lines) TAF

Results Messages

	book_id	title	author	category	price	publication_year	copies
1	109	SQL Cookbook	Anthony Molinaro	Programming	925.75	2026	8
2	119	Machine Learning	Tom Mitchell	Programming	1983.75	2026	35
3	120	AI Fundamentals	Andrew Ng	Programming	2380.50	2025	27
4	117	SQL in 10 Minutes	Ben Forta	Programming	899.30	2025	6
5	106	Database Systems	Elmasri	Programming	1587.00	2025	15
6	105	Data Structures	Narasimha Karumanchi	Programming	1190.25	2024	30
7	116	Clean Code	Robert C. Martin	Programming	1256.38	2024	20
8	113	Atomic Habits	James Clear	Self Help	800.00	2023	28
9	102	Python Basics	John Smith	Programming	991.88	2023	18
10	101	Learning SQL	Alan Beaulieu	Programming	859.63	2022	25
11	110	HTML & CSS	Jon Duckett	Programming	793.50	2022	22
12	114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17
13	115	Think and Grow Rich	Napoleon Hill	Finance	500.00	2021	9
14	112	Wings of Fire	A.P.J Abdul Kalam	Biography	350.00	2021	40
15	107	Operating Systems	Galvin	Programming	1256.38	2021	10
16	103	Java Complete Guide	Herbert Schildt	Programming	1124.13	2021	12
17	104	C Programming	Dennis Ritchie	Programming	727.38	2020	5

Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Co

Add publisher column

SQLQuery5.s...deepa (77)) * SQL project 5...A\deepa (75))

```
87      when copies>=20 then 'Available'
88      when copies>=5 then 'Limited'
89      else 'Out of Stock'
90      end as Copies_Current_Status
91      from books;
92
93      select * from books order by publication_year desc;
94
95      alter table books add publisher varchar(20);
96
97      select * from books;
```

100 % No issues found Ln: 97, Ch: 1 (20 chars, 1 lines) T

Results Messages

	book_id	title	author	category	price	publication_year	copies	publisher
1	101	Learning SQL	Alan Beaulieu	Programming	859.63	2022	25	NULL
2	102	Python Basics	John Smith	Programming	991.88	2023	18	NULL
3	103	Java Complete Guide	Herbert Schildt	Programming	1124.13	2021	12	NULL
4	104	C Programming	Dennis Ritchie	Programming	727.38	2020	5	NULL
5	105	Data Structures	Narasimha Karumanchi	Programming	1190.25	2024	30	NULL
6	106	Database Systems	Elmasri	Programming	1587.00	2025	15	NULL
7	107	Operating Systems	Galvin	Programming	1256.38	2021	10	NULL
8	109	SQL Cookbook	Anthony Molinaro	Programming	925.75	2026	8	NULL
9	110	HTML & CSS	Jon Duckett	Programming	793.50	2022	22	NULL
10	111	The Alchemist	Paulo Coelho	Novel	450.00	2020	14	NULL
11	112	Wings of Fire	A.P.J Abdul Kalam	Biography	350.00	2021	40	NULL
12	113	Atomic Habits	James Clear	Self Help	800.00	2023	28	NULL
13	114	Rich Dad Poor Dad	Robert Kiyosaki	Finance	650.00	2022	17	NULL
14	115	Think and Grow Rich	Napoleon Hill	Finance	500.00	2021	9	NULL
15	116	Clean Code	Robert C. Martin	Programming	1256.38	2024	20	NULL
16	117	SQL in 10 Minutes	Ben Forta	Programming	899.30	2025	6	NULL
17	119	Machine Learning	Tom Mitchell	Programming	1082.75	2026	26	NULL

Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Col: 1

Select Repository

Drop Members table

```
89      else 'Out of Stock'
90      end as Copies_Current_Status
91      from books;
92
93      select * from books order by publication_year desc;
94
95      alter table books add publisher varchar(20);
96
97      select * from books;
98
99      Drop table members;
```

100 % No issues found Ln: 99, Ch: 1 (19 chars, 1 lines) TABS LF

Messages

Commands completed successfully.

Completion time: 2026-08-02T20:18:51.8686001+05:30

100 % No issues found Ln: 4, Ch: 1 TABS MIXED UTF-8 wit

Query executed successfully. DEEPA\ (17.0 RTM) DEEPA\deepa (77) credo 00:00:00 Row: 1, Col: 1 0

Select Repository